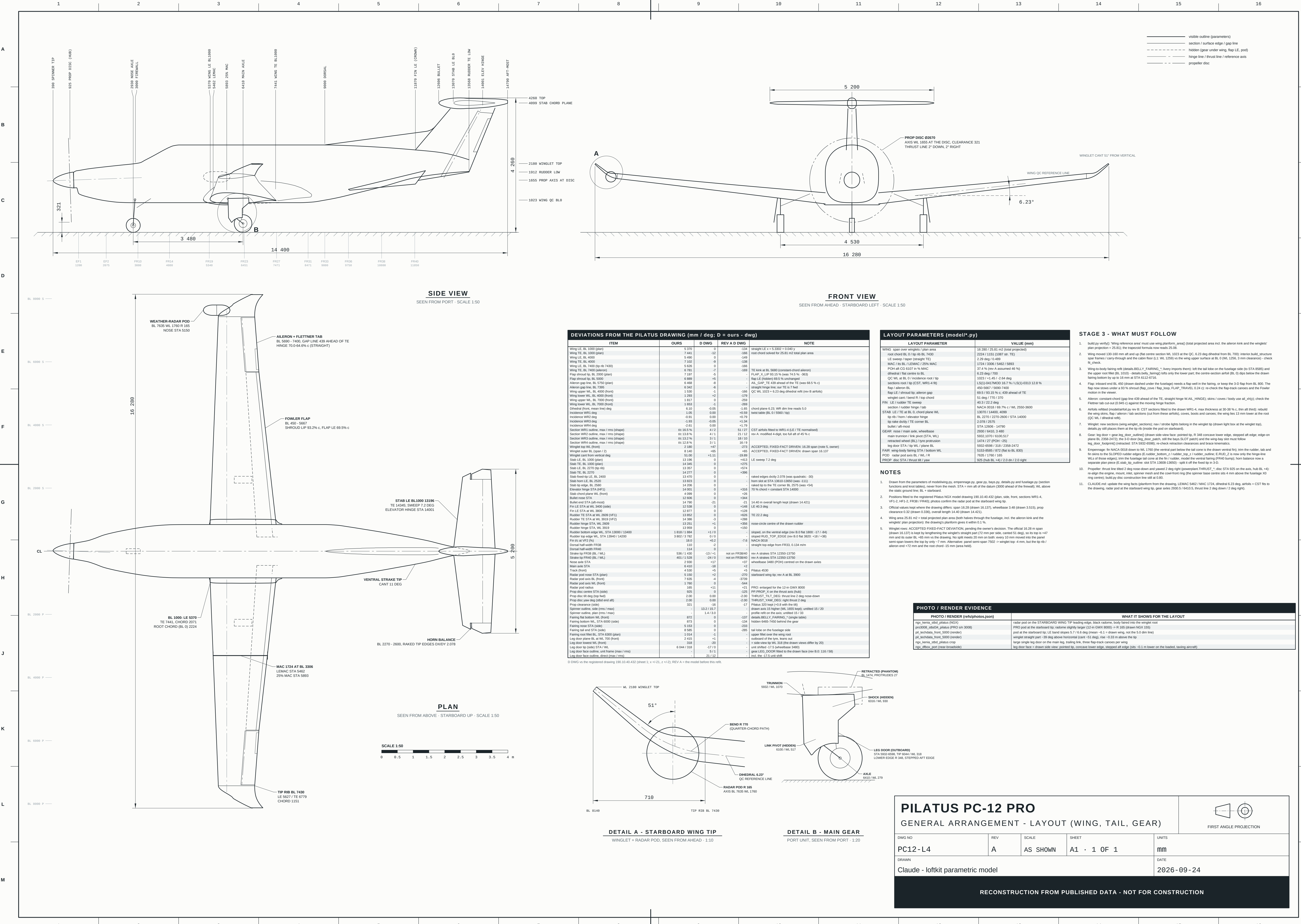




DEVIATIONS FROM THE PILATUS DRAWING (mm / deg; D = ours - dwg)					NOTE
ITEM	OURS	D DWG	REV A D DWG		
Wing LE, BL 1000 (plan)	5 370	0	-134	straight LE x = 5,3302 ± 0.040 y	
Wing TE, BL 1000 (plan)	7 441	-12	-166	root chord solved for 25.81 m2 total plan area	
Wing LE, BL 4000	5 490	0	-149		
Wing LE, BL 4000	7 102	-9	-138		
Wing LE, BL 7400 (tip rib 7430)	5 626	0	-166		
Wing TE, BL 7400 (airfoil)	6 781	-7	-169	TE kink at BL 5680 (constant-chord airfoil)	
Flap shroud lip, BL 2000 (plan)	7 187	-5		- FLAP_X_LIP 93.15 % (was 74.5 % -36C)	
Flap shroud lip, BL 5000	6 689	+5		- flap LE (inboard) 93.5 % unchanged	
Aileron gap line, BL 5750 (plan)	6 468	-8		- AIL_GAP: TE 439 ahead of the TE (was 68.5 % c)	
Aileron gap line, BL 7395	6 342	-6		- straight hinge line: our TE is 7 fwd	
Wing upper WL, BL 4000 (front)	1 530	-1	-166	QC WL 1023 ± 6.23 deg dihedral left (rev B airfoil)	
Wing lower WL, BL 4000 (front)	1 293	+2	-179		
Wing upper WL, BL 7000 (front)	1 817	0	-259		
Wing lower WL, BL 7000 (front)	1 653	-1	-269		
Dihedral (horiz. mean line) deg	6.10	-0.05	-1.65	root plane 6.23; WL din line reads 5.0	
Incidence WR1 deg	1.05	0.00	+0.39	heel table (BL 0 / 5560 / tip)	
Incidence WR2 deg	-0.91	0.00	+0.79		
Incidence WR3 deg	-1.93	-0.05	+1.34		
Incidence WR4 deg	-2.61	0.00	+1.79		
Section WR1 outline, max / rms (shape)	16.18 %	4 / 2	31 / 27	CST airfoils fitted to WR1 & LE / TE normalised	
Section WR2 outline, max / rms (shape)	16.13 %	4 / 1	21 / 12	rev A, modified 4 deg; too full aft of 45 ° c	
Section WR3 outline, max / rms (shape)	16.12 %	3 / 1	18 / 10		
Section WR4 outline, max / rms (shape)	16.12 %	3 / 1	18 / 10		
Winglet top WL (front)	2 180	+47		ACCEPTED, FIXED-FACT DRIVEN: 16.28 span (note 5, owner)	
Winglet outer BL (open / 2)	8 140	+65		ACCEPTED, FIXED-FACT DRIVEN: drawn span 16.137	
Winglet cant from vertical deg	51.00	+1.11	-19.89		
Shab BL, BL 1000 (plan)	13 196	0	+413	LE sweep 7.2 deg	
Shab TE, BL 1000 (plan)	14 345	0	+275		
Shab LE, BL 2270 (tip rib)	13 357	0	+574		
Shab TE, BL 2270	14 277	0	+396		
Shab head/rib LE, BL 2400	13 470	0		- raked edges dxdy 2.078 (was quadratic -30)	
Shab horn LE, BL 2520	13 823	0		- horn slot at STA 13630-13650 (was -131)	
Shab tip edge, BL 2580	14 206	0		- raked tip to the TE corner BL 2575 (was +54)	
Elevator hinge STA (HF1)	14 001	0	+316	70 % chord = constant STA 14000	
Sub chord plane WL (front)	4 099	0	+26		
Bullet nose STA	12 006	0	+26		
Bullet end STA (aft-most)	14 790	-21	-21	14.40 m overall length kept (drawn 14.421)	
Fin LE STA at WL 3000 (plan)	12 538	+148			
Fin LE STA at WL 3800	12 877	0	+128		
Rudder TE STA at WL 2609 (VF1)	13 862	0	+626	TE 22.2 deg	
Rudder TE STA at WL 3910 (VF2)	14 386	-3	+288		
Rudder hinge STA, WL 2609	13 251	+1	+356	nose-circle centre of the drawn rudder	
Rudder hinge STA, WL 3910	13 569	0	+150		
Rudder bottom edge WL, STA 13000 / 13400	1 818 / 1 884	+1 / 0		- sloped, on the ventral edge (rev B.0 flat 3800: -17 / -84)	
Rudder top edge WL, STA 13940 / 14200	3 802 / 3 782	0 / 0		- sloped RUDD_TOP_EDGE (rev B.0 flat 3800: +18 / +38)	
Fin tip VF1 (fin)	81.0	+62.2	-7.6	NACA 0018	
Dorsal half-width FR38	110	-2		- straight top edge from FR33, 0.134 mm	
Dorsal half-width FR40	114	0			
Strike tip FR38 (BL / WL)	536 / 1 430	-13 / +1		rev A strikes STA 12350-13750	
Strike tip FR40 (BL / WL)	401 / 1 528	-24 / 0		rev A strikes STA 12250-13750	
Nose axle STA	2 930	+17		wheelbase 3460 (POH) centred on the drawn axles	
Main axle STA	6 410	-18	+2		
Track (front)	4 530	+5	+5	Pilatus 4530	
Radar pod nose STA (plan)	5 150	+2	-270	starboard wing lip; rev A at BL 3900	
Radar pod axle BL (front)	7 630	-4	-3739		
Radar pod axle WL (front)	1 760	-4	-344		
Radar pod radius	165	+11	+21	PRO: enlarged for the 12-in GWX 8000	
Prop disc centre STA (side)	905	0	-125	PRO: PROPC_X on the thrust axis (fin)	
Prop disc tilt deg (top fwd)	2.00	0.00	-2.00	THRUST_TILT_DEG: right thrust 2 deg nose-down	
Prop clearance (side)	2.00	0.00	-2.00	THRUST_YAW_DEG: right thrust 2 deg	
Spinner outline, side (rms / max)	321	-16		Pilatus 320 kept (+18 with the H)	
Spinner outline, plan (rms / max)	-	13.2 / 15.7		drawn axle 15 higher (WL 1655 kept; untiled 15 / 20)	
Fairing flat bottom WL (front)	872	0	-137	details: BELLY_FAIRING_* (single table)	
Fairing bottom WL, STA 6000 (side)	873	0	-134	radon 6465-7450 behind the gear	
Fairing nose STA (side)	5 153	0	-253		
Fairing tail end STA (side)	8 585	0	-285	tail tube on the fuselage side	
Fairing root fillet BL, STA 6300 (plan)	1 054	-1		- upper fillet over the wing root	
Leg door lower WL (front)	2 415	+1		- outboard of the tyre, leans out	
Leg door top (side) STA / WL	6 044 / 318	-17 / 0		- side view tip WL 318 (the drawn views differ by 20)	
Leg door face outline, unit frame (rms)	-	5 / 1		- unit offset: 17.5 (wheelbase 3460)	
Leg door face outline, direct (max / rms)	-	21 / 12		- gear LEG_DOOR fitted to the drawn face (rev B.0: 116 / 58)	

D DWG vs the registered drawing 190.10.40.432 (sheet 1, x=+21, z=+2); REV A = the model before this edit.

LAYOUT PARAMETERS (model/*_py)		VALUE (mm)
LAYOUT PARAMETER		
WING span over wingtips / plan area		16 280 / 25.81 m2 (total projected)
root chord BL 0 / tip rib BL 7430		2224 / 1151 (1087 se: TE)
LE sweep / taper (straight TE)		2.29 deg / 0.449
MAC / fin BL / LEMAC / 25% MAC		1224 / 5300 / 5462 / 5893
POH aft CG 107 % MAC		37.4 % (rev A assumed 46 %)
FLAP_X_LIP 93.15 % (was 74.5 % -36C)		6.23 deg / 700
QC WL at BL 0 / incidence root / tip		1023 / +1.45 / -2.64 deg
sections root / tip (CST, WR1-4 fin)		LSQ(1.0417MOD 16.7 % / LSQ(1.0313 12.8 %)
flap / aileron BL		450-5667 / 5690-7400
flap LE / shroud lip; aileron gap		68.5 / 93.15 % c 450 ahead of TE
winglet cant / bend R / top chord		51 deg / 770 / 370
FIN LE / rudder TE sweep		40.3 / 22.2 deg
section / rudder hinge / tab		NACA 0018 / 69.7 % c / WL 2550-3600
STAB LE / TE at BL 0, chord plane WL		13070 / 14400, 4099
tip rib / horn / elevator hinge		BL 2270 / 2270-2600 / STA 14000
tip rake dxdy / TE corner BL		2.078 / 2575
buffer / aft-most		STA 12606 - 14790
GEAR nose / main axle, wheelbase		2930 / 6410, 3 490
main transon / line pivot (STA, WL)		5932 / 1071 / 6100.517
retracted wheel (BL) / tyre protrusion		1474 / 27 (POH -25)
leg door STA / tip WL / plane BL		5932-6598 / 318 / 2358-2472
FAIR wing/body fairing STA / bottom WL		5153-5895 / 872 (But to BL 830)
POD radar pod axle BL / WL / R		7035 / 1760 / 165
PROPC disc STA / thrust BL / yaw		925 (hub BL -40) / 2.0 dh / 2.0 right

NOTES

- Drawn from the parameters of modelling wing py, empennage py, gear py, bays py, details py and fuselage py (section functions and knot tables), never from the mesh. STA = mm aft of the datum (3000 ahead of the firewall); WL above the static ground line. BL = starboard.
- Positions fitted to the registered Pilatus NGX model drawing 190.10.40.432 (sheet 1, side, front, sections WR1-4, VF1-2, HF1-2, FR38 / FR40), photos confirm the radar pod at the starboard wing tip.
- Critical values kept where the drawing differs: span 16.28 (drawn 16.137), wheelbase 3.48 (drawn 3.515), prop clearance 0.32 (drawn 0.355), overall length 14.40 (drawn 14.421).
- Wing area 25.81 m2 = total projected plan area (both halves through the fuselage, incl. the aileron kink and the wingtip plan projection); the drawing's platform gives it within 0.1 %.
- Winglet new: ACCEPTED FIXED-FACT DEVIATION, pending the owner's decision. The official 16.28 m span (drawn 15.137) is kept by lengthening the winglet's straight part (72 mm per side, cantled 51 deg), so its top is +47 mm and its outer BL +45 mm vs the drawing. No split meets 20 mm on both: every 10 mm moved into the panel semi-open lowers the top by only ~7 mm. Alternative: panel semi-open 7502 > winglet top -4 mm, but the tip rib / aileron end +72 mm and the root chord -15 mm (area held).

PHOTO / RENDER EVIDENCE		WHAT IT SHOWS FOR THE LAYOUT
PHOTO / RENDER (refs/photos/son)		
ngx_kernl_38td_pilatus (NGX)		radar pod on the STARBOARD WING TIP leading edge, black radome, body lared into the winglet root
pro008_38tdk_glass (PRO sh 3038)		PRO pod at the starboard tip; radome slightly longer (12-in GWX 8000) - R 165 (drawn NGX 155)
pl_schdata_front_3000 (render)		pod at the starboard tip; LE band slopes 5.7 / 6.6 deg (mean -6.1 - drawn wing root the 5.0 dim line)
pl_schdata_front_3000 (render)		winglet straight part -39 deg above horizontal (cant -51 deg), rise -0.33 m above the tip
ngx_kernl_38td_pilatus (ngx)		large single leg door on the main leg, trailing fin, three flap-track canoes per wing
ngx_athos_port (near-boardside)		leg door face = drawn side view; pointed tip, concave lower edge, stepped aft edge (axis -0.1 m lower on the loaded, taxiing aircraft)

PILATUS PC-12 PRO

GENERAL ARRANGEMENT - LAYOUT (WING, TAIL, GEAR)

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L4	A	AS SHOWN	A1 · 1 OF 1	mm
DRAWN				DATE
Claude - loftkit parametric model				2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION